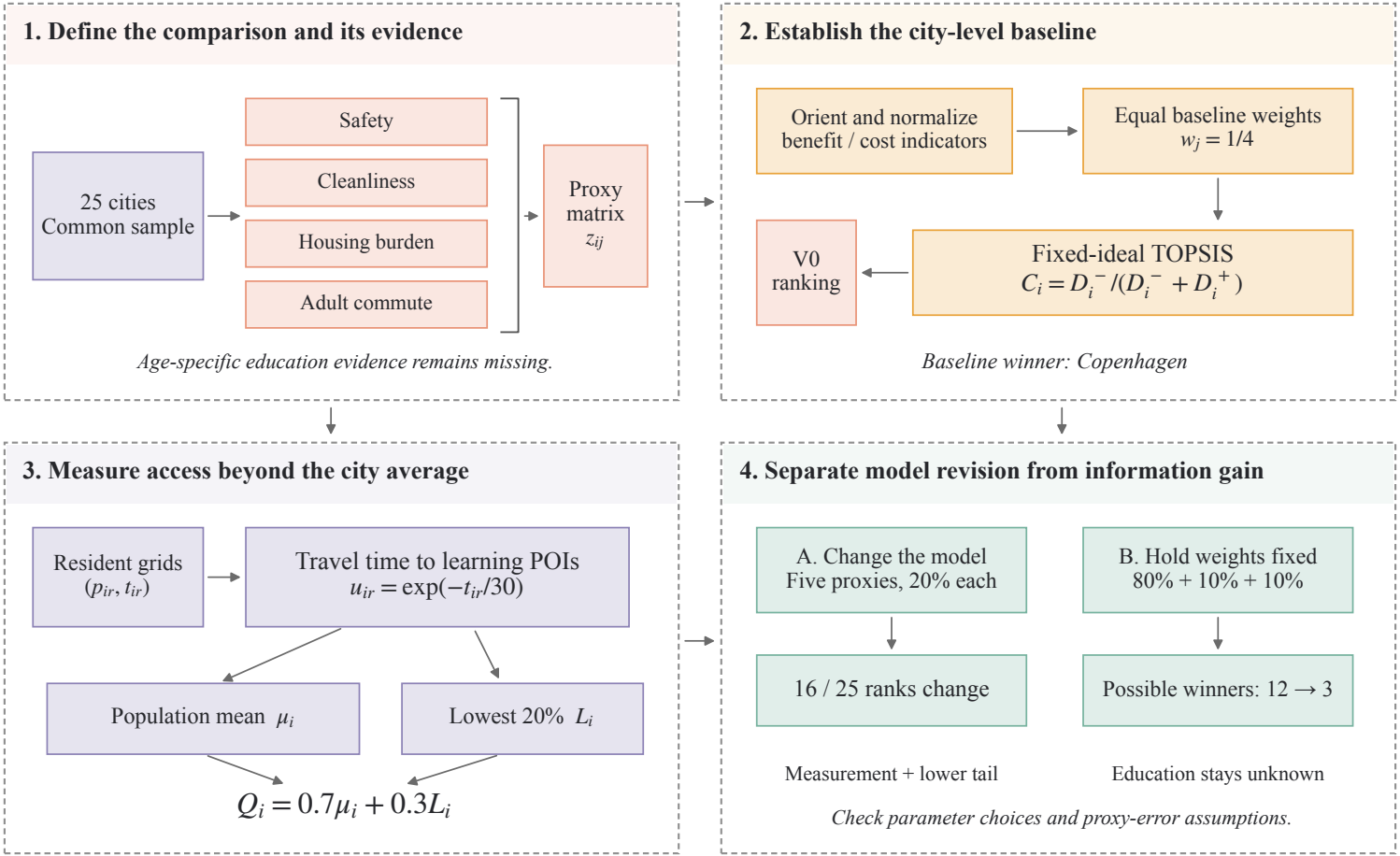
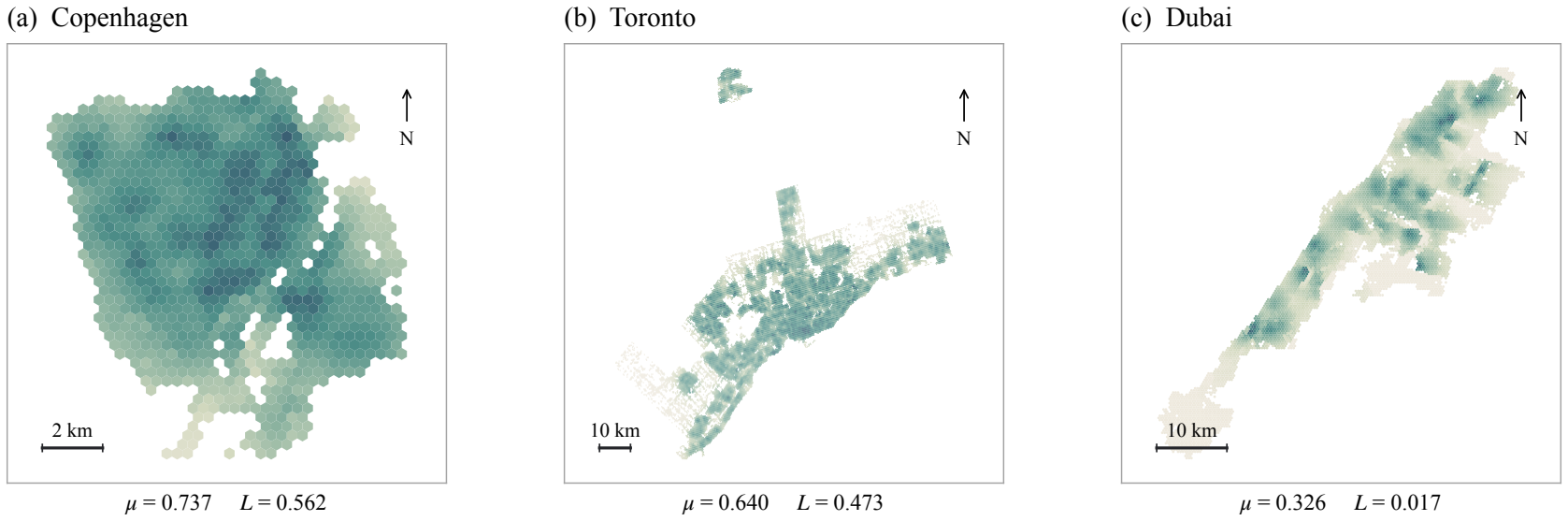
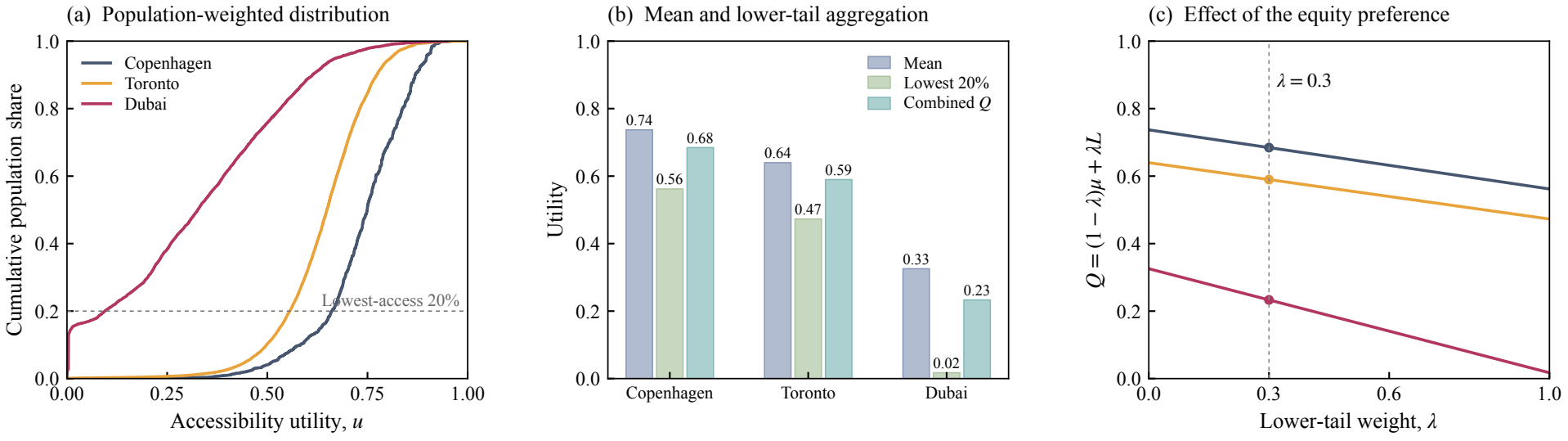

1. Define the comparison and its evidence
25 cities
Common sample
Safety
Cleanliness
Housing burden
Adult commute
Proxy matrix
z_{ij}
Age-specific education evidence remains missing.
2. Establish the city-level baseline
Orient and normalize benefit / cost indicators
Equal baseline weights
$w_j = 1/4$
Fixed-ideal TOPSIS
$C_i = D_i^- / (D_i^- + D_i^+)$
V0 ranking
Baseline winner: Copenhagen
3. Measure access beyond the city average
Resident grids
(p_{ir}, t_{ir})
Travel time to learning POIs
$u_{ir} = \exp(-t_{ir}/30)$
Population mean μ_i
Lowest 20% L_i
$Q_i = 0.7\mu_i + 0.3L_i$
4. Separate model revision from information gain
A. Change the model
Five proxies, 20% each
16 / 25 ranks change
Measurement + lower tail
B. Hold weights fixed
80% + 10% + 10%
Possible winners: 12 → 3
Education stays unknown
Check parameter choices and proxy-error assumptions.

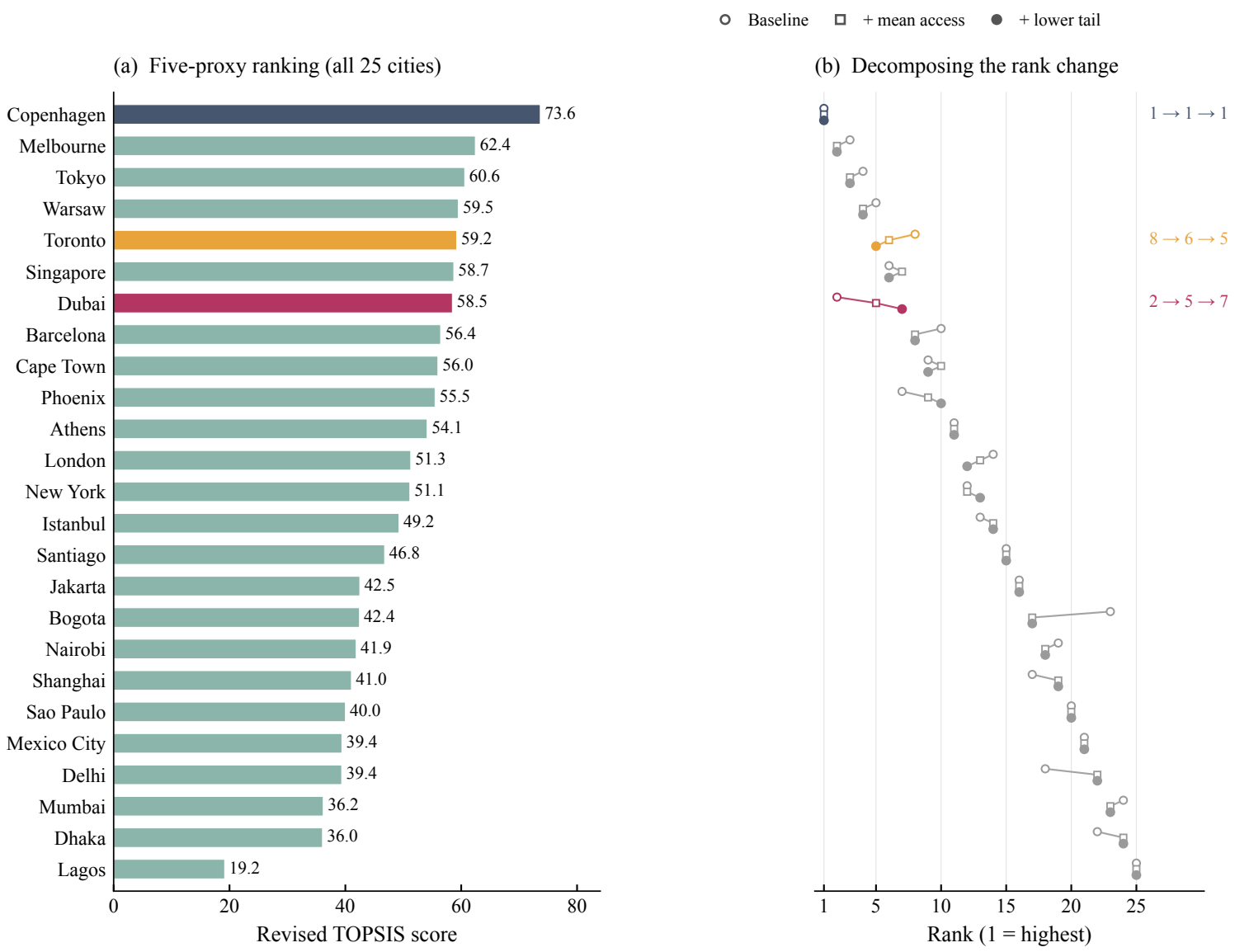


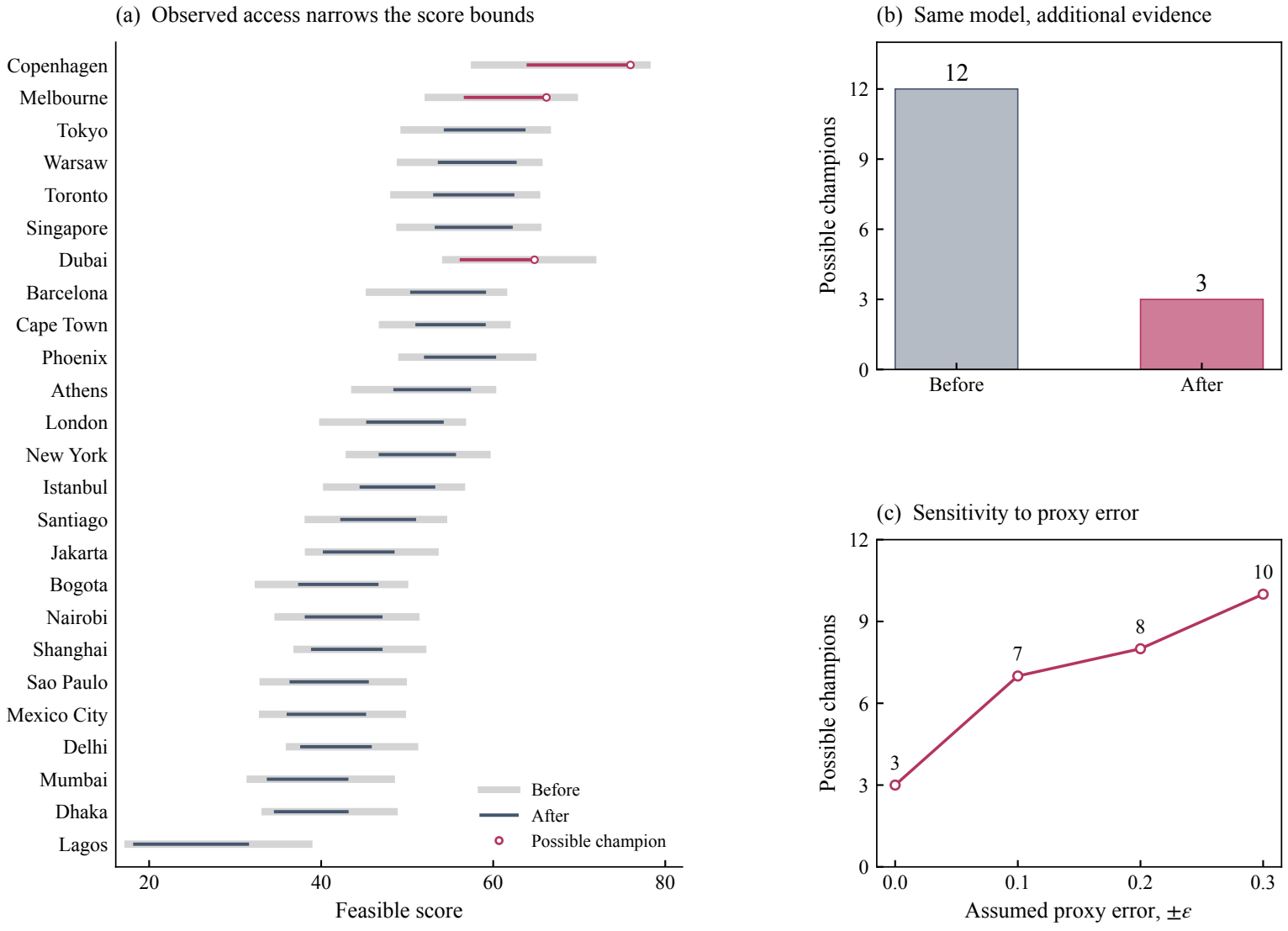
$\mu = 0.737$ $L = 0.562$

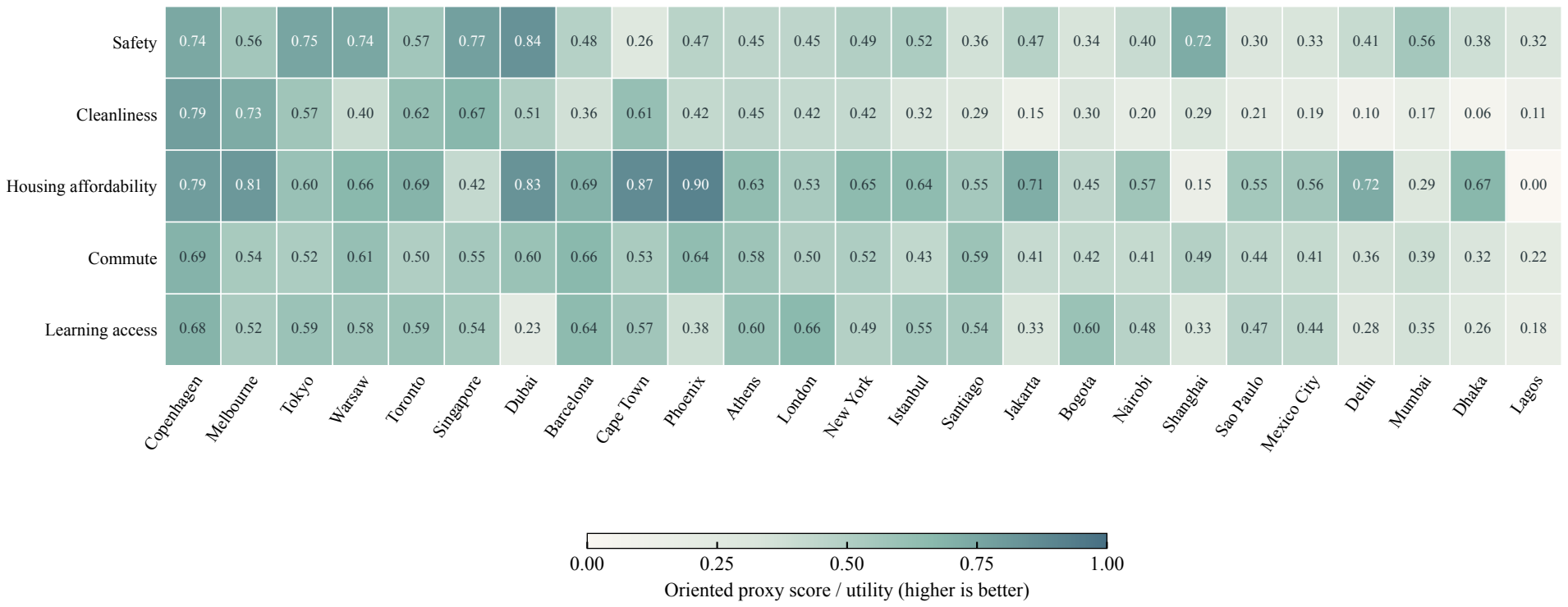
$\mu = 0.640$ $L = 0.473$

$\mu = 0.326$ $L = 0.017$


(a) Population-weighted distribution
Copenhagen
Toronto
Dubai
Lowest-access 20%
Cumulative population share
Accessibility utility, u
0.0
0.2
0.4
0.6
0.8
1.0
0.00
0.25
0.50
0.75
1.00
(b) Mean and lower-tail aggregation
Mean
Lowest 20%
Combined Q
Utility
0.74
0.56
0.68
0.64
0.47
0.59
0.33
0.02
0.23
Copenhagen
Toronto
Dubai
(c) Effect of the equity preference
λ = 0.3
Q = (1 − λ)μ + λL
Lower-tail weight, λ
0.0
0.3
0.6
1.0


Baseline
+ mean access
+ lower tail
(a) Five-proxy ranking (all 25 cities)
Copenhagen 73.6
Melbourne 62.4
Tokyo 60.6
Warsaw 59.5
Toronto 59.2
Singapore 58.7
Dubai 58.5
Barcelona 56.4
Cape Town 56.0
Phoenix 55.5
Athens 54.1
London 51.3
New York 51.1
Istanbul 49.2
Santiago 46.8
Jakarta 42.5
Bogota 42.4
Nairobi 41.9
Shanghai 41.0
Sao Paulo 40.0
Mexico City 39.4
Delhi 39.4
Mumbai 36.2
Dhaka 36.0
Lagos 19.2
0
20
40
60
80
Revised TOPSIS score
(b) Decomposing the rank change
1 → 1 → 1
8 → 6 → 5
2 → 5 → 7
1
5
10
15
20
25
Rank (1 = highest)


(a) Observed access narrows the score bounds
Copenhagen
Melbourne
Tokyo
Warsaw
Toronto
Singapore
Dubai
Barcelona
Cape Town
Phoenix
Athens
London
New York
Istanbul
Santiago
Jakarta
Bogota
Nairobi
Shanghai
Sao Paulo
Mexico City
Delhi
Mumbai
Dhaka
Lagos
Before
After
Possible champion
20
40
60
80
Feasible score
(b) Same model, additional evidence
Possible champions
0
3
6
9
12
12
3
Before
After
(c) Sensitivity to proxy error
Possible champions
0
3
6
9
12
3
7
8
10
0.0
0.1
0.2
0.3
Assumed proxy error, ±ε


Safety 0.74 0.56 0.75 0.74 0.57 0.77 0.84 0.48 0.26 0.47 0.45 0.45 0.49 0.52 0.36 0.47 0.34 0.40 0.72 0.30 0.33 0.41 0.56 0.38 0.32
Cleanliness 0.79 0.73 0.57 0.40 0.62 0.67 0.51 0.36 0.61 0.42 0.45 0.42 0.42 0.32 0.29 0.15 0.30 0.20 0.29 0.21 0.19 0.10 0.17 0.06 0.11
Housing affordability 0.79 0.81 0.60 0.66 0.69 0.42 0.83 0.69 0.87 0.90 0.63 0.53 0.65 0.64 0.55 0.71 0.45 0.57 0.15 0.55 0.56 0.72 0.29 0.67 0.00
Commute 0.69 0.54 0.52 0.61 0.50 0.55 0.60 0.66 0.53 0.64 0.58 0.50 0.52 0.43 0.59 0.41 0.42 0.41 0.49 0.44 0.41 0.36 0.39 0.32 0.22
Learning access 0.68 0.52 0.59 0.58 0.59 0.54 0.23 0.64 0.57 0.38 0.60 0.66 0.49 0.55 0.54 0.33 0.60 0.48 0.33 0.47 0.44 0.28 0.35 0.26 0.18
Copenhagen
Melbourne
Tokyo
Warsaw
Toronto
Singapore
Dubai
Barcelona
Cape Town
Phoenix
Athens
London
New York
Istanbul
Santiago
Jakarta
Bogota
Nairobi
Shanghai
Sao Paulo
Mexico City
Delhi
Mumbai
Dhaka
Lagos
0.00
0.25
0.50
0.75
1.00
Oriented proxy score / utility (higher is better)